Stage 3 Price / Premium Surrogate Report
Direct Value Versus Positive Continuation-Premium Prediction

Codex-assisted surrogate modelling review

June 19, 2026

Abstract

This report reviews the Stage 3 surrogate experiment on the v1 small-grid American option dataset. Two fixed multilayer perceptrons were compared: a direct value surrogate and a positive-premium surrogate that predicts continuation premium and reconstructs value as payoff plus premium. The experiment uses the existing v1 dataset read-only, keeps Delta, Gamma, and boundary fields as diagnostics only, does not save model weights, and does not train boundary or Greek heads. The Stage 3 review decision is `READY_FOR_BOUNDARY_DIAGNOSTIC_STAGE`.

Contents

1 Purpose of Stage 3

Stage 3 asks a narrow modelling question: does predicting nonnegative continuation premium help preserve the American payoff obstacle better than predicting option value directly? The experiment is intentionally small and fixed. It is not an architecture search, not a production-risk model, and not a final research conclusion.

2 Link to v1 Dataset QA

The source dataset is `results/04_surrogate_dataset/v1_small_grid/dataset_v1_small_grid.npz`. The v1 QA report accepted 288 regimes and 1,498,464 sampled rows for price-surrogate planning. The solver source remains the validated baseline American CN/PSOR solver, grounded in the finite difference American option formulation used throughout the validation ladder (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

Split	Rows	Role
Train	1,051,006	Training split; preprocessing fit only here.
Validation	98,857	Regime-level validation split.
Test	223,729	Regime-level test split.
Stress holdout	124,872	Held-out stress combinations.

3 Model Families

Direct value model. The direct baseline uses the seven normalized v1 features and predicts `value_over_K` with a linear output layer.

Positive-premium model. The premium model uses the same seven features plus `payoff_over_K`, predicts `premium_over_K`, and applies a Softplus output so the predicted premium is nonnegative. The value prediction is reconstructed as

$$\widehat{U}/K = \Phi/K + (\widehat{U - \Phi})/K.$$

The payoff is analytic contract information computed from spot, strike, and option type. It is not target leakage: it is known before pricing and is used only for reconstruction and obstacle-aware structure.

4 Target Policy

The only training targets are `value_over_K` and `premium_over_K`. The fields `boundary_spot_over_K`, `delta`, and `scaled_gamma` remain diagnostic only. No boundary head, Delta head, Gamma head, or production Greek target is trained in this stage.

5 Preprocessing Policy

Both models use `StandardScaler` fit on training rows only. Validation, test, and stress holdout rows do not influence scaling. The train subset is capped deterministically:

$$\text{seed} = 20260619, \quad \text{cap} = 250,000, \quad \text{selected rows} = 250,000.$$

If the train split exceeds the cap, rows are selected with `np.random.default_rng(20260619).choice(train_indices,replace=False)` and sorted before batching.

6 Loss Functions and Training

Both MLPs use mean squared error and Adam with learning rate $10^{-}3$. Training used 10 epochs and batch size 8192. No model weights or `.pt` files were saved.

Model	Final train loss	Final validation loss
Direct value MLP	2.558e-04	2.742e-04
Positive-premium MLP	4.389e-05	2.359e-05

7 Evaluation Protocol

Metrics are reported by regime-level split and by diagnostic region. Relative errors use a stable denominator floor of $10^{-}4$. Obstacle and negative-premium rates use a near-zero rate tolerance of $10^{-}4$, with violation amounts counted only above $10^{-}10$.

8 Results by Split

Model	Split	Value RMSE	Value MAE	Premium RMSE
direct_value_mlp	validation	0.016558	0.013540	0.016558
direct_value_mlp	test	0.033351	0.023965	0.033351
direct_value_mlp	stress_holdout	0.028907	0.023260	0.028907
positive_premium_mlp	validation	0.004857	0.003073	0.004857
positive_premium_mlp	test	0.012269	0.007788	0.012269
positive_premium_mlp	stress_holdout	0.009516	0.006607	0.009516

9 Direct Value Versus Positive Premium

The direct model is the plain baseline. The positive-premium model adds American-option structure by learning the continuation premium and reconstructing value from the analytic payoff. This comparison therefore tests whether the obstacle-aware representation reduces payoff-obstacle violations while keeping price accuracy close to the direct value baseline.

10 Obstacle and Negative-Premium Diagnostics

Split	Obstacle rate	Negative premium rate	Review
train	0	0	PASS
validation	0	0	PASS
test	0	0	PASS
stress_holdout	0	0	PASS

11 Near-Boundary and Strict-Interior Diagnostics

Boundary-near rows use the v1 diagnostic mask inherited from Ticket 09/10 logic. Strict-interior rows exclude maturity, payoff-kink, boundary-near, and nonfinite Greek regions. These regions are not separate training targets; they are evaluation slices.

Model	Split	Region	Rows	Value RMSE
direct_value_mlp	test	boundary_near	2,219	0.013360
direct_value_mlp	test	strict_interior	193,872	0.034586
direct_value_mlp	stress_holdout	boundary_near	6,341	0.023537
direct_value_mlp	stress_holdout	strict_interior	103,124	0.029066
positive_premium_mlp	test	boundary_near	2,219	0.008631
positive_premium_mlp	test	strict_interior	193,872	0.011874
positive_premium_mlp	stress_holdout	boundary_near	6,341	0.002972
positive_premium_mlp	stress_holdout	strict_interior	103,124	0.008976

12 Figures

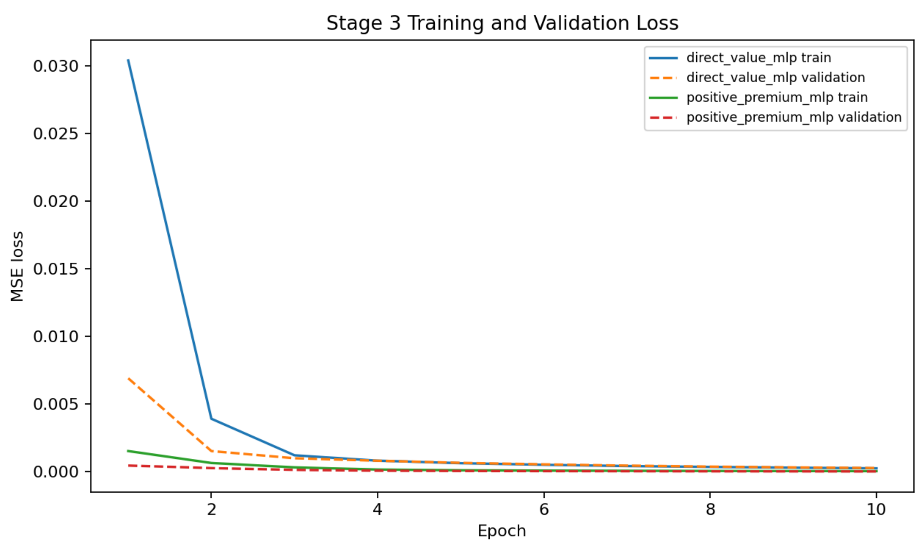

Figure 1: Loss curves.

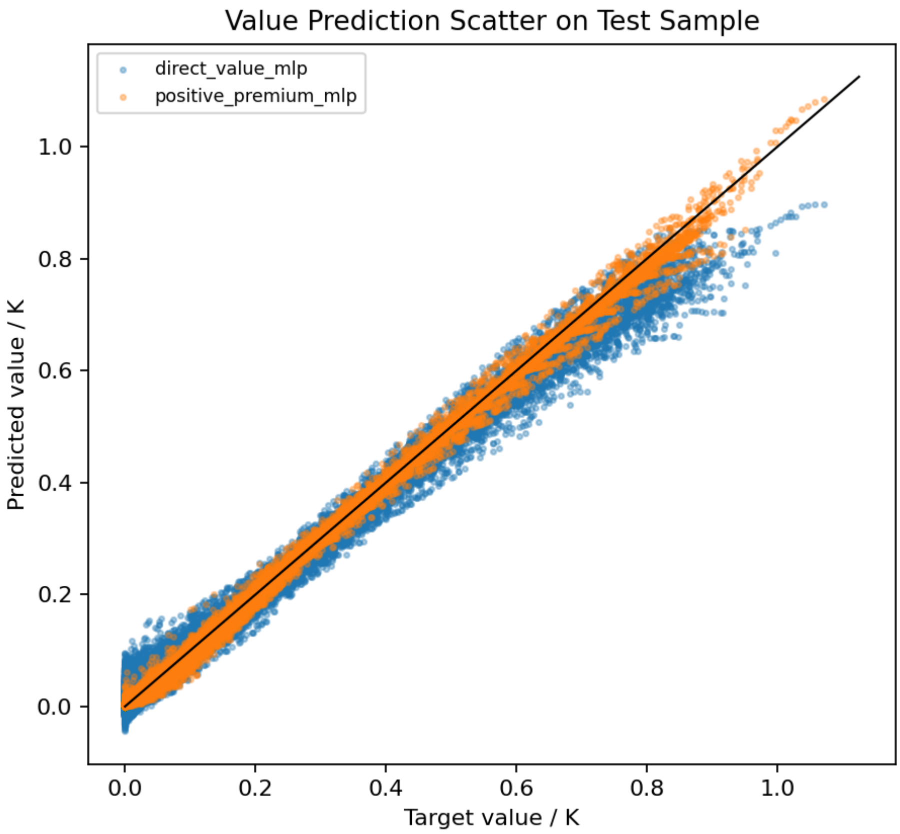

Figure 2: Value prediction scatter.

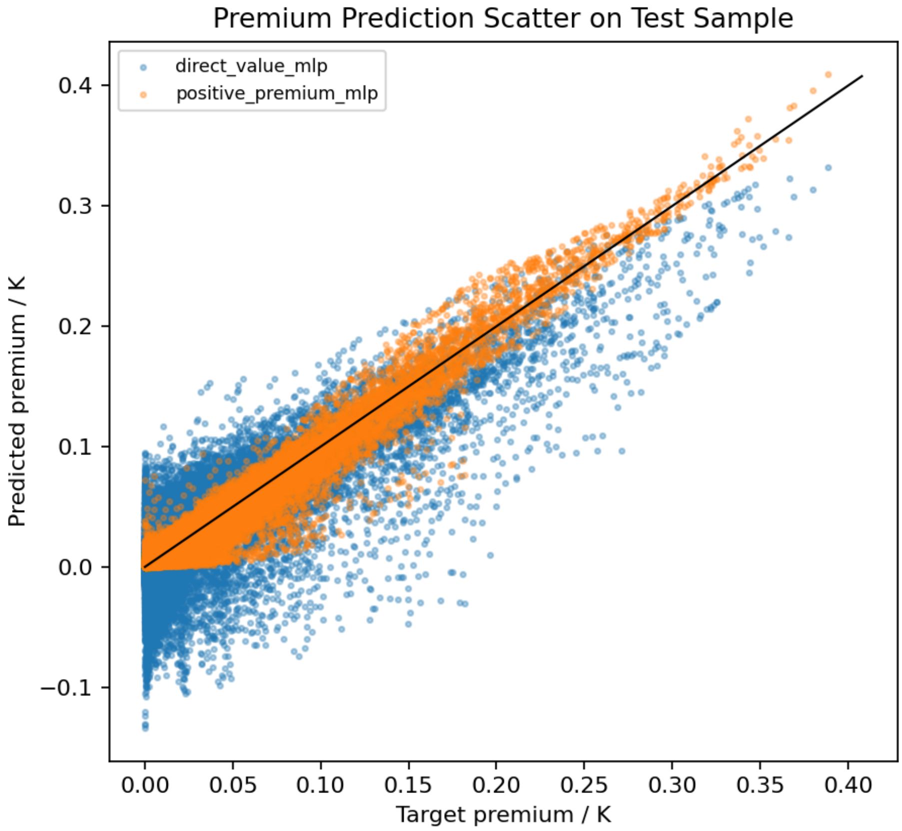

Figure 3: Premium prediction scatter.

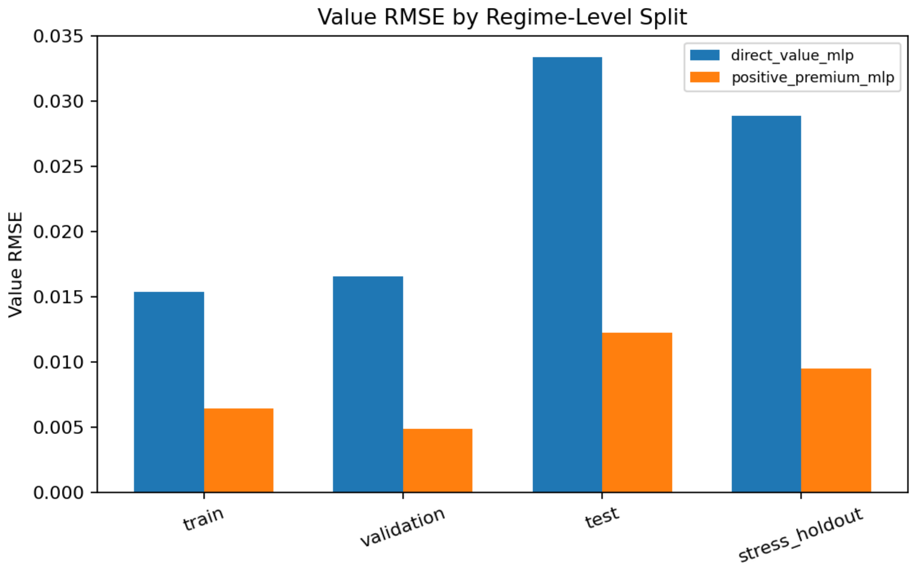

Figure 4: Error by split.

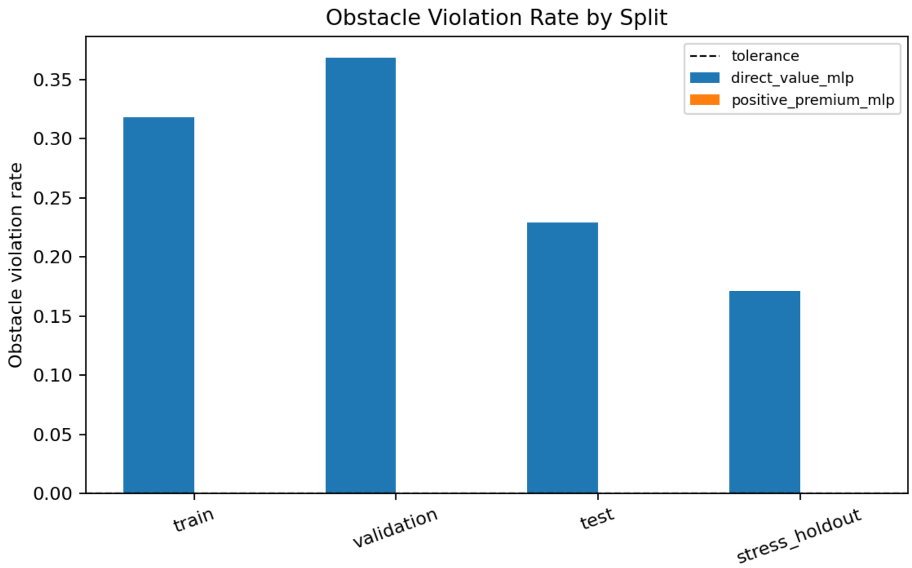

Figure 5: Obstacle violation comparison.

13 Limitations

This experiment uses one fixed MLP configuration and a capped training subset. It does not search architectures, tune hyperparameters broadly, train a boundary head, train Greek heads, or save model weights. Delta and Gamma remain diagnostic fields. The stress-holdout results are informative but not a production-risk guarantee.

14 What Stage 3 Supports

Stage 3 supports a first comparison between direct value prediction and an obstacle-aware positive-premium representation. It also verifies that preprocessing can respect regime-level split separation and that metrics can be reported by split, boundary-near region, and strict-interior region.

15 What Stage 3 Does Not Support

This stage does not approve production deployment, production Greek labels, broad architecture selection, boundary-head training, or final paper claims. It does not modify the validated solver and does not regenerate the v1 dataset.

16 Recommended Next Stage

If the review decision is `READY_FOR_BOUNDARY_DIAGNOSTIC_STAGE`, the next stage may plan a boundary-diagnostic modelling experiment. If the decision is `REVIEW_REQUIRED_BEFORE_BOUNDARY_STAGE`, the next step is human review of the price/premium errors, obstacle diagnostics, and training setup before adding any new target heads.

References

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